

HARSH GUPTA

harsh160502@gmail.com | +91 9982346893 | [Github-Harsh1652](#) | [Linkedin-Harsh-gupta16](#)

Summary

Backend Engineer with 1.5+ years building production systems for healthcare and SaaS, specializing in AI application engineering RAG and multi-agent systems with LangGraph shipped to production. Built a call-intelligence platform processing 200+ calls/day that cut median lead response time from 4 hours to under 8 minutes, and an attribution platform providing single-dashboard visibility into which channels and campaigns drive revenue. Own backend features end-to-end, from design through production.

Technical Skills

Languages: JavaScript, TypeScript, Python, SQL

Backend & APIs: Node.js, Express.js, FastAPI, REST API design, SSE (streaming APIs), JWT auth, RBAC, multi-tenant architecture

Databases & Caching: PostgreSQL, MongoDB, Redis, Prisma ORM, Pinecone (vector database)

AI/LLM Engineering: RAG (hybrid retrieval: BM25 + dense vectors + reranking), LangGraph agents, OpenAI API, Cohere, embeddings, LangSmith tracing/observability, prompt caching

Infrastructure & Tools: AWS (EC2), Docker, CI/CD, Git, n8n, WhatsApp Business API, speech-to-text APIs

Experience

Reinvent Digital — healthcare marketing platform

Gen AI Engineer (Backend)

Oct 2025 – Present

- Designed and shipped an automated lead-response and call-intelligence pipeline (Node.js, WhatsApp Business API, Sarvam speech-to-text, LLMs, n8n) processing 200+ calls/day; cut median lead response time from 4 hours to under 8 minutes.
- Built an attribution platform giving the business single-dashboard visibility into which advertising channels and campaigns are driving revenue showing where to focus spend and how to optimize each campaign.
- Built a programmatic SEO (P-SEO) platform that automated local content generation and publishing across service × city combinations, integrating LLM-based content generation, quality validation, and WordPress publishing workflows.

Excollo — software services / SaaS products

Software Engineer — AI-Persona (multi-tenant conversational AI SaaS)

Apr 2025 – Oct 2025

- Designed and built the platform backend (Node.js, Express, PostgreSQL, Prisma) with multi-tenant architecture, tenant isolation, RBAC, and audit logging; deployed in production for one client and architected for easy onboarding of future tenants.
- Built an automated document-ingestion pipeline (parsing, chunking, OpenAI embeddings, Pinecone indexing) that replaced manual lookups across documents, meeting notes, and transcripts cutting time to find information from a few hours to a few minutes..
- Designed a hybrid retrieval pipeline (BM25 keyword search + dense vector search + Cohere reranking) across tenant-specific knowledge bases to improve answer relevance.

Backend Developer Intern — CRUDO (order management system)

Feb 2025 – Mar 2025

- Built the backend for a multi-tenant order management system (Node.js, MongoDB) serving 100+ shop owners, with ERP integration, Redis-backed sessions, and Pinecone-based customer segmentation; transitioned to a full-time Software Engineer role.

Projects

Multi-Agent Contract Analysis System — LangGraph, GPT-4o, Python

Apr 2026

- Built a multi-agent document-intelligence backend orchestrating specialized analysis and review agents with LangGraph, streaming results over SSE with structured, Pydantic-validated outputs.
- Added LangSmith tracing and prompt caching, cutting LLM inference cost by 50%.

Stock Research Agent — LangGraph, FastAPI, Python

Feb 2026

- Built a 5-agent stock research system with LangGraph (parallel fan-out across news, financials, and sentiment agents) generating Buy/Hold/Sell reports for US and Indian markets in under 30 seconds.
- Implemented state merging and aggregation across parallel agents with LangGraph reducers, and instrumented full observability via LangSmith, tracking per-agent latency and cost.

Regulatory Document Intelligence Platform (RHP/DRHP) — Node.js, MongoDB, Pinecone

Oct 2025 – Jan 2026

- Built a RAG backend for large regulatory filings: recursive chunking, embeddings, vector retrieval, Cohere reranking, and structured prompting for summarization and cross-document comparison.
- Designed the multi-tenant service around it — workspace isolation, RBAC, JWT authentication — with ingestion and comparison workflows automated via n8n.

Education

MCA, UPES 2023 – 2025

BCA, St. Xavier's College 2020 – 2023